6/11/2023
FP_TD6

Q1) let x = 10 in x+1

$$\text{(LET)}\ \frac{\text{VAL}\ \frac{}{10 \Downarrow 10} \qquad \text{ADD}\ \frac{\text{VAL}\ \frac{}{10 \Downarrow 10} \quad \text{VAL}\ \frac{}{1 \Downarrow 1} \quad 11 = 10+1}{10+1 \Downarrow 11}}{\text{let } x = 10 \text{ in } x+1 \Downarrow 11}$$

Q1) let x = 10 in let y = 1 in x+y

$$\text{LET}\ \frac{\text{VAL}\ \frac{}{10 \Downarrow 10} \qquad \text{LET}\ \frac{\text{VAL}\ \frac{}{1 \Downarrow 1} \quad \text{ADD}\ \frac{\text{VAL}\ \frac{}{10 \Downarrow 10} \quad \text{VAL}\ \frac{}{1 \Downarrow 1} \quad 11 = 10+1}{10+1 \Downarrow 11}}{\text{let } y = 1 \text{ in } 10+y \Downarrow 11}}{\text{let } x = 10 \text{ in } (\text{let } y = 1 \text{ in } x+y) \Downarrow 11}$$

Q2) $\frac{\text{VAL}}{\text{true} \Downarrow \text{true}}$ $\frac{\text{VAL}}{\text{false} \Downarrow \text{false}}$ $\text{NOT1}\ \frac{b \Downarrow \text{true}}{\text{not } b \Downarrow \text{false}}$ $\text{NOT2}\ \frac{b \Downarrow \text{false}}{\text{not } b \Downarrow \text{true}}$

$\&\&_1\ \frac{b_1 \Downarrow \text{false}}{b_1 \,\&\&\, b_2 \Downarrow \text{false}}$ $\&\&_2\ \frac{b_1 \,\&\, \text{true} \quad b_2 \Downarrow v}{b_1 \,\&\&\, b_2 \Downarrow v}$

$\|_1\ \frac{b_1 \Downarrow \text{true}}{b_1 \,\|\, b_2 \Downarrow \text{true}}$ $\|_2\ \frac{b_1 \Downarrow \text{false} \quad b_2 \Downarrow v}{b_1 \,\|\, b_2 \Downarrow v}$

Q3) $v ::= x \mid \overset{S}{\text{fun}}\,(x:\tau) \Rightarrow e$

$\frac{\text{VAL}}{v \Downarrow v}$ $\text{APP}\ \frac{e_2 \Downarrow v \quad e_1 \Downarrow \lambda x.e \quad e\{v/x\} \Downarrow v'}{e_1 \,\#\, e_2 \Downarrow v'}$

$\lambda x.e \;\; v \xrightarrow[\beta]{} e\{v/x\} \xrightarrow{\#} v'$

$\text{CONCAT}\ \frac{e_1 \Downarrow s_1 \quad e_2 \Downarrow s_2 \quad s = s_1 \wedge s_2}{e_1 \wedge e_2 \Downarrow s}$ $\frac{e \Downarrow v}{\text{raise } e \Downarrow \text{raise } v}\ \text{RAISE}\ v$

$\frac{e \Downarrow \text{raise } v}{\text{raise } e \Downarrow \text{raise } v}$ (raise raise)

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$$\frac{e_2 \Downarrow \text{raise } v}{e_1\ e_2 \Downarrow \text{raise } v} \text{ (raise APPd)}$$

$$\frac{e_2 \Downarrow v_2 \quad e_1 \Downarrow \text{raise } v_1}{e_1\ e_2 \Downarrow \text{raise } v_1} \text{ (raise APPg)}$$

$$\frac{e_1 \Downarrow v_1}{\text{try } e_1 \text{ with } x.e_2 \Downarrow v_1} \text{ (TRYV)}$$

$$\frac{e_1 \Downarrow \text{raise } v_1 \quad e\{v_1/x\} \Downarrow v}{\text{try } e_1 \text{ with } x.e_2 \Downarrow v} \text{ (TRYR)}$$

$$\frac{e_1 \Downarrow \text{raise } v_1 \quad e_2\{v_1/x\} \Downarrow \text{raise } v}{\text{raise } e_1 \text{ with } x.e_2 \Downarrow \text{raise } v} \text{ (TRYR raise)}$$

Q4) se restreindre au cas $e \to^* v$ (plusieurs étapes de réduction)
$e \to^* \text{raise } v$

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